

Sai Siddartha Alleni

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PROFESSIONAL SUMMARY

Robotics Software Engineer and M.S. candidate at Purdue University specializing in reinforcement learning for legged locomotion, robot perception, and autonomous navigation. Experienced in training parallelized RL policies (2,000+ environments) using Isaac Lab and PyTorch, with a history deploying ROS2-based SLAM and multi-robot coordination layers. Seeking Robotics Software or ML Engineering roles focused on sim-to-real transfer, computer vision, and robot learning.

TECHNICAL SKILLS

- **Core AI/ML:** PyTorch, Stable Baselines3 (SB3), Reinforcement Learning (PPO, SAC), Deep Learning, Vision Transformers, Diffusion Models
- **Robotics Frameworks:** ROS/ROS2, Isaac Lab, MuJoCo, Gazebo, Nav2, SLAM Toolbox, MoveIt2
- **Perception & Vision:** YOLOv11, RGB-D Processing, Point Cloud Processing (PCL), Depth Estimation, Camera Calibration, PCA Pose Estimation
- **Programming:** Python, C++, MATLAB/Simulink, CUDA, Bash
- **DevOps & Tools:** Git, Docker, Linux, JIRA, Scrum/Agile, CI/CD

PROJECTS

- **Reinforcement Learning for Quadruped Locomotion** [G](#) Oct 2025 – Present
Isaac Lab, PyTorch, PPO, Domain Randomization, LiDAR
 - Trained an **asymmetric actor-critic PPO policy** across 2,000 parallel simulation environments, enabling Unitree Go2 blind locomotion across 7 terrain types with 90% velocity tracking accuracy.
 - Designed a **LiDAR-based observation space** with proprioceptive state history for robust walking, reducing velocity command error to ± 0.1 m/s across randomized terrain friction and motor dynamics.
 - Applied **domain randomization** (terrain geometry, body mass, joint damping) to improve policy generalization for physical transfer.
 - Optimized training using **GPU-parallelized simulation** to achieve 3x faster iteration cycles over single-env baselines.
- **Depth Estimation with Vision Transformers & Diffusion Models** Jan 2026 – Mar 2026
PyTorch, Vision Transformers, Diffusion Policy, RGB-D
 - Implemented a **diffusion-based monocular depth estimation** pipeline inspired by Marigold, utilizing a Vision Transformer encoder for dense RGB-to-depth prediction on custom robotics datasets.
 - Evaluated the deployment of **diffusion policies** with depth prediction models for end-to-end robotic manipulation tasks.
- **Autonomous Navigation & SLAM System** Sept 2024 – Dec 2024
ROS2, Nav2, SLAM Toolbox, LiDAR, Gazebo
 - Developed a full **ROS2 Navigation2 stack** with SLAM Toolbox for real-time occupancy grid mapping and A* path planning.
 - Configured a **behavior tree-based recovery system** with a DWB local planner for dynamic obstacle avoidance at 200Hz.
 - Integrated **LiDAR scan matching with EKF sensor fusion** for robust localization in feature-sparse landscapes.
- **Multi-Robot Perception & Coordination System** Jan 2025 – May 2025
ROS, YOLOv11, Modbus TCP, RGB-D, Node-RED
 - Built a **real-time perception-to-grasp pipeline** using YOLOv11 and RGB-D camera calibration to achieve sub-2cm pose accuracy.
 - Architected a **Modbus TCP communication layer** between a Fetch AGV and Techman cobot using an event-driven state machine.

EXPERIENCE

- **Visteon Corporation** July 2023 – July 2024
Software Engineer Bangalore, India
 - Designed a **Python-based build analysis tool** used by 5 teams, reducing project onboarding time via automated dependency graphs.
 - Maintained a **Jenkins and JFrog CI/CD pipeline** for continuous integration of automotive middleware across QNX platforms.
 - Recovered **3% CPU utilization** on resource-constrained QNX 8.0 targets by finding and removing trailing production debug blocks.
 - Validated **DDS middleware communication** for autonomous systems via Wireshark analysis and Scapy-based test automation.
- **Visteon Corporation** Jan 2023 – June 2023
Software Engineering Intern Bangalore, India
 - Developed **11 custom Clang-Tidy safety checks** (MISRA C:2012 rules) for QNX RTOS instrument cluster firmware.
 - Built a **CMake-integrated static analysis pipeline** with the LLVM toolchain, enabling real-time feedback inside VS Code.

EDUCATION

- **Purdue University** July 2024 – May 2026
Master of Engineering in Robotics West Lafayette, IN
- **Vellore Institute of Technology** July 2019 – June 2023
Bachelor of Technology in Electronics and Communication Vellore, India